

Group 3 - Space Complexity

Members:

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User Evaluation Report

Methods for user evaluation

We recruited 7 participants from our ENG1 cohort, as they represent the target audience for the game. We managed to recruit our participants by randomly finding people during one of our scheduled sessions within the lab and asking whether they were able to test our game. This ensured relevance, availability, and diversity within a shared university context, as we gathered people of different genders, ages and experiences to test our game and give their opinions. Each participant signed an informed consent form to meet ethical standards. We used the following tools for efficient data collection:

- Google Forms: To gather structured feedback and task completion data.
- Observation Notes: To document user actions, errors, and hesitations.

These tools were chosen for their accessibility, ease of use, and ability to provide both quantitative and qualitative data. They offered the participants ways of giving written feedback or rating their experiences on a scale, with the user being given a laptop if they did not have one available to complete their feedback.

The procedure we followed for the user evaluation was the following:

1. Introduction: Explained the evaluation process and emphasized that the system, not the user, was being tested.
2. Tasks: Participants completed four tasks (placing buildings, handling events, and achieving satisfaction goals).
3. Think-Aloud Protocol: Users verbalized their thoughts while performing tasks, helping us understand their mental models.
4. Observation: We noted user actions, delays, and errors using a predefined template.
5. Post-Task Questionnaire: Collected feedback on ease of use, confusion points, and improvement suggestions via Google Forms.
6. Rationale: This method provided comprehensive insights through:
 - Think-Aloud: Understanding user thought processes.
 - Observation: Identifying real-time issues.
 - Google Forms: Structured feedback collection.

Severity Scale for the usability table:

- **Critical:** Prevents task completion.
- **Major:** Significantly hinders task performance.
- **Minor:** Causes inconvenience or confusion but doesn't prevent task completion.
- **Cosmetic:** Aesthetic issue with minimal impact on usability.

Usability problems found by users

Problem ID	Description	Severity	Frequency	User Comments	Proposed Solution
P1	Difficulty understanding how satisfaction is calculated	Major	4/7 users	"How do you reach a higher score?" and "The satisfaction number is really confusing."	Change the satisfaction score to a percentage number, where 100% is clearly the maximum and 0% is the minimum.
P2	Difficulty distinguishin	Major	3/7 users	"Wasn't sure which building	Add labels to the sprites depending on

	g between building types.			was for studying or eating.”	the category of buildings (Accommodation, Lecture Hall, etc)
P3	Lack of visible pause button	Critical	4/7 users	“Pause is not very clear.” and “Add a pause button.”	Add a visible pause button to the game or specify that to pause the button you have to press the esc key.
P4	Achievements are unclear and difficult to understand.	Minor	4/7 users	“Didn’t know what the badges were for.” and “Achievements were confusing.”	Add descriptions for achievements, a label for the badges specifying that those are the achievements and instructions on how to achieve them.
P5	Accidental deletion of buildings due to unclear control/ Not knowing how to demolish a building	Minor	2/7 users	“Right-clicking deletes a building by accident.” and “How do you demolish a building?”	Add a button for demolishing buildings and a confirmation prompt for building deletion.
P6	Ending menu is easy to accidentally exit the game	Minor	1/7	“The menu popped up, and I accidentally exited it.”	Require name input before allowing exit of the game.
P7	Difficulty balancing resources and understanding building costs and differences between buildings of the same category	Major	2/7	“Balance is not very clear, not sure how much buildings cost” and “I’m not sure the difference between buildings of the same category”	Display building costs and descriptions more prominently
P8	No instructions displayed at the start of the game	Critical	5/7 users	“Instructions at the start would make the game easier to understand”	Add a tutorial or instructions screen at the start of the game.